FrameCrafter: Novel View Synthesis as Video Completion

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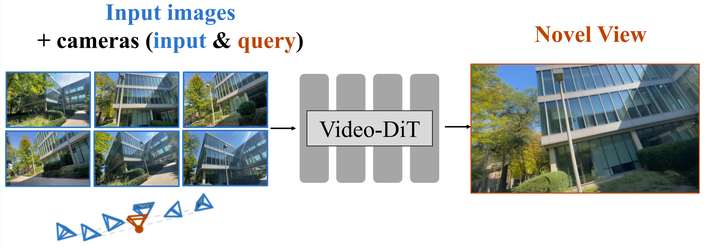
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Project Page
Code Available

Make video diffusion "forget" about time for sparse-view NVS: turning a pretrained video model into a multi-view model with a strong prior.

Problem Setup & Motivation

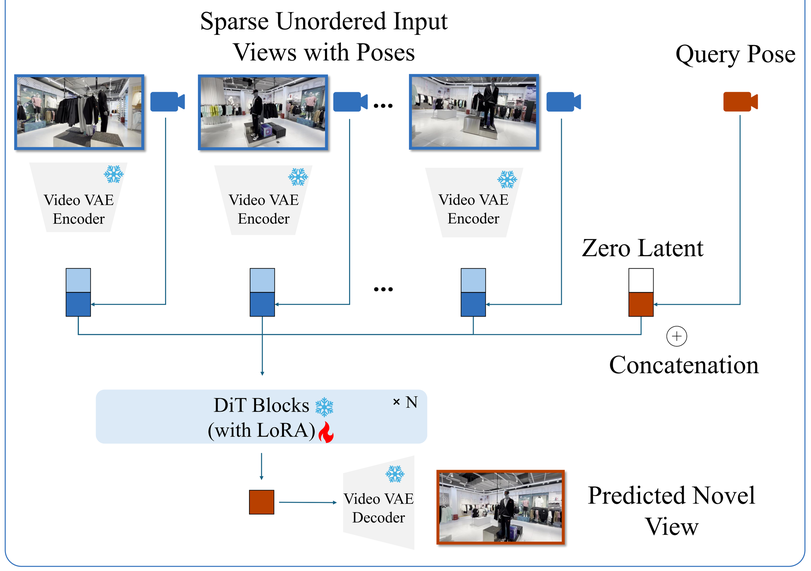


Key thesis: video diffusion models provide a stronger prior for sparse-view NVS than image diffusion models.

Prior Methods (Image Prior)	FrameCrafter (Video Prior)
✗ Weak multi-view prior	✓ Strong inherent multi-view prior
✗ Requires large-scale multi-view data	✓ Lightweight finetuning with limited data

Challenge: Sparse views are **unordered** → the video model must "forget" about time.

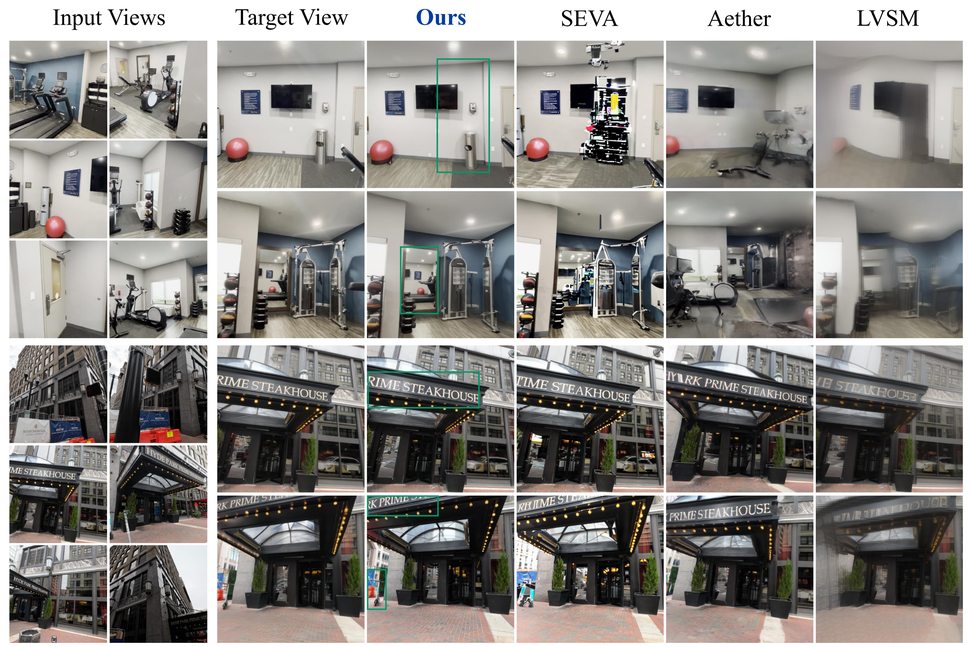
Method



Results

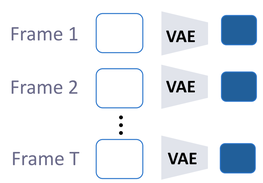
Method	Backbone	#Scenes	DL3DV		Mip360	
			PSNR↑	LPIPS↓	PSNR↑	LPIPS↓
EscherNet	SD1.5	10K	12.07	0.484	11.14	0.540
Aether	CogVideoX-5B	-	12.66	0.469	12.60	0.651
SEVA	SD2.1	80K	16.15	0.253	14.59	0.372
Ours	CogVideoX-5B	1K	13.65	0.389	13.09	0.540
Ours	Wan2.1-14B	1K	17.18	0.223	15.64	0.365

6-view NVS. Best results are highlighted as first , second , third.



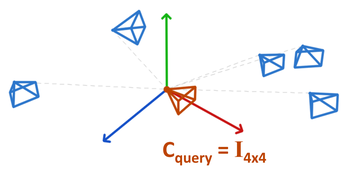
Three Simple Changes: Video Model → Multi-View Model

1 Per-View Independent VAE Encoding



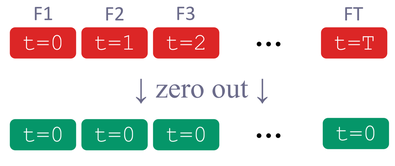
Each view → 1-frame "video"

2 Query-centered Camera Conditioning



All poses relative to query camera

3 Remove Temporal Positional Encoding



No frame ordering bias

Ablations & Analysis

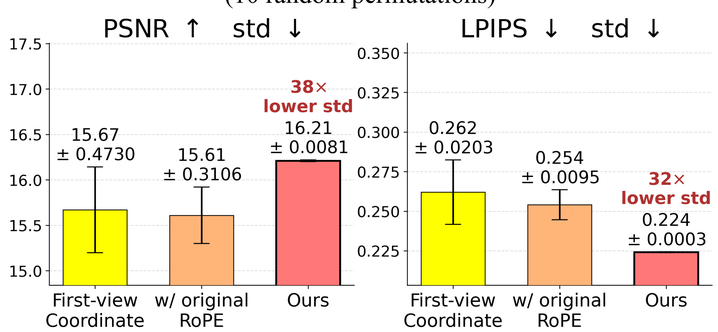
Ablations

Variant	PSNR↑	LPIPS↓
✗ Per-View Encoding	11.50	0.676
✗ Pixel Unshuffle	13.53	0.359
✗ Target-Only Supervision	15.08	0.270
w/ PRoPE	14.91	0.315
w/ Original RoPE	15.30	0.265
Full model	15.69	0.246

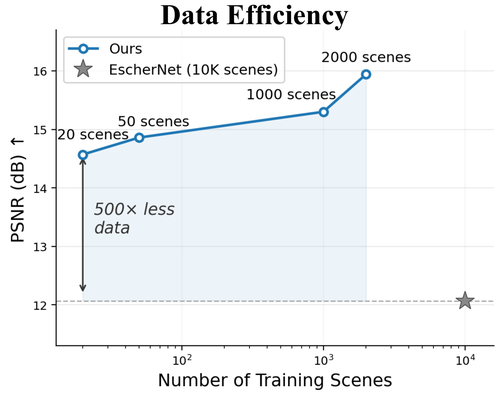
192×336 6-view NVS on DL3DV

Permutation Invariance

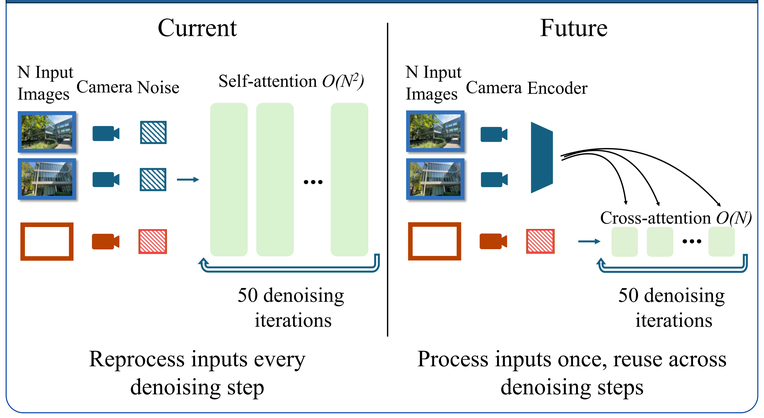
(10 random permutations)



Data Efficiency



Future Work



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